YOUR GCP PORTFOLIO / EXPLAINED SIMPLY

From blood-cell data to trusted answers

The complete PBMC + Open Knowledge Format project plan

Imagine a small data factory: measurements arrive, an inspector checks them, workers organize the results, and a helpful guide explains what the numbers mean.

2,700	2,638	62
cells checked	cells pass QC	cells fail QC

Where you are: Steps 1-3 are complete. Step 4 has working cloud QC and a verified workflow. BigQuery/dbt cloud automation is next. Steps 5-7 cover OKF, the assistant, and the portfolio demonstration.

Your project: pbmc-okf-portfolio
Cloud region: europe-west1
Guide date: 01 November 2026

Working = verified today. Partly done = working components awaiting integration. Planned = a proposed next component, not a deployed resource.

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The whole project in one picture

Follow the arrows from measurements to an answer. The words in each box describe its job.



The supporting tools

GitHub code > Docker package > Artifact Registry shelf > Cloud Run

Terraform = blueprints | IAM = keys | Cloud Logging = run diary

PARTLY DONE means it works from your computer; cloud automation is next.

The full count matrix stays in Storage. BigQuery holds metadata and QC tables.

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What is inside the box?

PBMC means peripheral blood mononuclear cells: a group of blood cells used in this public dataset.

Think of a spreadsheet

Imagine one row for each cell and one column for each gene. A number in the grid records that gene's measured count in that cell. Most boxes contain zero, so we store only the useful entries rather than wasting space on every zero.

Item	In this project	Simple meaning
Input	Public 10x PBMC3k, distributed through Scanpy	One small, repeatable teaching dataset.
Size	2,700 cells and 32,738 gene features	The rows and columns we must preserve.
Source file	AnnData / H5AD	A box containing the count grid and labels.
Donors	One healthy donor	This is not a disease-comparison study.
Attribution	10x Genomics; CC BY 4.0	Keep the credit and source with the data.

QC is an inspector, not a trash bin

Quality control (QC) puts a pass/fail sticker on each cell. The demo policy requires at least 200 detected genes, fewer than 2,500 detected genes, and less than 5% mitochondrial counts. A zero-count cell always fails.

The measured result is **2,638 passing and 62 failing**: 5 cells exceed the gene threshold and 57 exceed the mitochondrial threshold. These rules are screening choices, not diagnoses or a proven doublet detector.

Nothing is silently thrown away
All 2,700 cells remain in the exported count matrix and QC table. The passing-cell cohort is a separate, clearly labelled view of the results.

Important: the starting file is already a filtered gene-barcode count matrix. This project does not process raw sequencing FASTQ files, infer cell types, or claim that one donor represents a population.

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Your cloud resources today

Each resource has one clear responsibility. These names match the project's configuration and verified runs.

Resource / actual name	Like a...	Function and current use
GCP project pbmc-okf-portfolio Region: europe-west1	Named workshop	Groups the project's services, permissions, quota and billing.
Pipeline bucket pbmc-okf-portfolio-data	Storeroom	Stores pinned raw input, verified processed releases, and per-run audit records. Private and versioned.
State bucket pbmc-okf-portfolio-tfstate	Builder's inventory	Stores Terraform's record of existing infrastructure. Separate from scientific data; private and versioned.
Artifact Registry pbmc-containers	Package shelf	Stores Docker images. Immutable tags prevent replacing a tagged image; deployment pins an exact digest.
Cloud Run Job pbmc-qc	Temporary worker	Runs cloud QC using 1 vCPU and 2 GiB memory. One task; 900-second attempt limit; one retry allowed.
Workflows pbmc-qc-workflow	Factory manager	Starts the QC job, waits for completion, and returns its result. No timer or schedule is configured.
BigQuery pbmc_raw pbmc_analytics_r_<release_id>	Organized filing room	Raw tables hold cell/gene metadata and summaries. Analytics contains six dbt models. Live manual runs passed; automated jobs are next.
Cloud Logging	Run diary	Records execution logs and structured events. GCS audit files provide a separate durable record.

Storage layout: **raw/<source_checksum>/** for input, **processed/releases/<release_id>/** for outputs, and **runs/** for audits. The count matrix remains in Storage; it is not loaded into BigQuery.

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Tools, keys and future workers

A tool runs work; an identity grants permission. Working tools and proposed additions are labelled separately below.

Tool / resource	What it does	Status
Python + scientific libraries	AnnData, NumPy, pandas, SciPy and PyArrow read sparse counts, calculate QC and write validated files.	WORKING
dbt	Runs six SQL models and 30 data tests. DuckDB provides a local SQL practice environment.	WORKING Cloud job next
Docker + Artifact Registry	Package code and dependencies, then store the version that Cloud Run executes.	WORKING
Terraform + gcloud + ADC	Describe infrastructure, operate GCP, and authenticate local libraries without checking credentials into Git.	WORKING
Git + GitHub + VS Code	Keep code history, review changes, and provide a reproducible portfolio repository.	WORKING
Cloud Run: load / dbt jobs	Proposed pbmc-warehouse-load and pbmc-dbt-build jobs; pass the exact QC release between them.	PLANNED
OKF builder + validator	Proposed pbmc-okf-build job writes and checks versioned knowledge notes; store bundles in GCS and examples in Git.	PLANNED
Assistant service + AI model	Proposed pbmc-assistant Cloud Run service, Python API/UI, and Vertex AI-hosted Gemini. Choose and configure the model during Step 6.	PLANNED Recommended
GitHub Actions + WIF	Test changes and deploy approved versions using short-lived cloud identity, without storing a long-lived service-account key.	PLANNED
Monitoring + budget alerts	Track failures, duration and cost. Verify alert setup and permissions during operations work; alerts are not a spending stop.	PLANNED Verification

Keys already created: pbmc-qc-runner can read/create pipeline objects. pbmc-workflow-runner can invoke/read the QC job. Future workers should receive only the access their own stage needs. **Optional later:** Cloud Scheduler for timed runs; Secret Manager only if a future integration requires a secret.

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The seven-stage project plan

Each stage leaves something you can show, test, and explain in an interview.

Stage / status	What we do	Finished when...
1. Inspect the data COMPLETE	Understand the source, dimensions, count matrix and scientific limits.	The dataset, licence, shape and source checksum are recorded.
2. Reproducible QC COMPLETE	Pin the input, apply a versioned policy, keep stable IDs, and validate exports.	A checked release can be reproduced or safely reused, with a separate run audit.
3. Warehouse + dbt COMPLETE	Load BigQuery tables and create useful analytics with SQL tests.	Remote rows match source tables; six models and 30 dbt tests pass.
4. Cloud automation IN PROGRESS	Run containerized stages using Terraform, IAM, GCS, Cloud Run and Workflows.	One workflow runs QC, the warehouse load and dbt for the same release, with safe recovery.
5. OKF knowledge PLANNED	Generate and validate a knowledge bundle explaining the data and its meaning.	Every generated note has a valid structure, correct links and traceable source/release context.
6. Scientist assistant PLANNED	Connect a simple interface to approved knowledge and read-only data tools.	Test questions return correct, cited answers and unsupported questions are handled honestly.
7. Operations + showcase PLANNED	Demonstrate failures, recovery, costs, CI/CD and a clean portfolio walkthrough.	A reviewer can understand, reproduce, test and watch the project end to end.

Today's checkpoint

Cloud QC and the first workflow are verified. The next work is connecting the already-working BigQuery loader and dbt project to that cloud workflow.

This is a small-data demonstration of reliable engineering. It does not establish large-scale performance, production availability, or scientific validity beyond the stated dataset and checks.

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Finish Step 4: four substeps

The warehouse logic already exists. We now give it cloud workers and connect the handoffs.

Substep	Implementation	Acceptance / rough effort
1. BigQuery load job	Download the exact verified GCS release. Run the existing loader in a container. Grant scoped input and warehouse permissions.	All three tables match their source data; a rerun adds no duplicate rows. **45-75 min**
2. dbt job	Package the dbt project and adapter. Configure the project, location and same release ID. Persist dbt results and logs.	Six models and all 30 tests pass in the release-specific dataset. **45-75 min**
3. Link the workflow	Run QC, then load, then dbt. Carry the explicit release ID and audit references through the chain. Stop if a stage fails.	One invocation completes every stage for the same release. **30-60 min**
4. Verify + document	Check safe replay, partial failure, missing/corrupt input, permissions, logs and cost evidence. Update the operating guide.	Tests and live evidence show correct success and recoverable failure. **30-60 min**

The most important handoff

A **release ID** is the name of one checked batch. Every stage must use that same batch name. Choosing whichever data happens to be called 'latest' can mix old QC with new analytics. Use an explicit, verified handoff instead.

Permissions are part of the design

Loading needs permission to run BigQuery jobs and write the intended tables. dbt needs read access to raw tables and write access to its analytics dataset. Creating a new per-release dataset needs additional permission; decide that explicitly rather than granting broad project-wide editing by default.

Time estimate
Allow roughly **3-5 hours across 2-3 guided sessions**, including integration and troubleshooting. This is a planning range, not measured execution time.

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OKF: the data's instruction book

Open Knowledge Format makes the meaning around data readable to both a person and an AI assistant.

A table can tell you that a value is 62. It may not tell you why those 62 cells failed, which rules were used, or whether they were removed. OKF notes explain that missing context. Think of a small folder of labelled study cards.

What the format is

The official OKF v0.2 specification uses Markdown documents with YAML metadata at the top. Documents can describe concepts and link to each other. The required concept metadata includes a type; other fields support discovery and context. This is a portable knowledge format, not a new database.

Project knowledge card	What our planned generator will explain
Dataset card	Source, attribution, dimensions, donor count, checksum and scientific boundaries.
Table cards	The meaning of columns, keys, joins and the associated BigQuery resource.
Metric cards	How counts, detected genes, mitochondrial percentage and pass fractions are computed.
QC policy card	Versioned thresholds, exclusion reasons, and why failed cells remain in storage.
Release / lineage card	Which source, code, runtime, policy and tested analytics belong to this release.
Playbook + question cards	Safe ways to inspect results, answer common questions and handle a failed run.

How we will know the notes are trustworthy

Generate measured facts from verified manifests and warehouse results. Validate metadata and links against a pinned specification version. Check referenced resources, release IDs and numeric claims. Publish the bundle only after its checks pass; keep changes visible in Git.

Example
Question: Were failed cells deleted?
Knowledge note: No. All cells remain in the count matrix and QC table; the passing-cell cohort is a separate selection.

Format reference: Official Open Knowledge Format specification. The proposed project card categories above are our design, not mandatory OKF document types.

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The assistant: a careful tour guide

Planned Step 6: a simple interface helps a scientist ask questions about this specific, verified dataset.

A question's journey

Step	What happens
1. Ask	The scientist asks: 'Why did 62 cells fail QC?'
2. Find context	Retrieve the matching release's dataset, policy and metric notes from the OKF bundle.
3. Get facts	Use a narrowly scoped read-only tool to fetch approved summary or exclusion counts from BigQuery.
4. Explain	A language model turns those facts into plain language. It must distinguish observed data from explanation.
5. Show evidence	Return the release ID, relevant knowledge references and the data/tool result used for the answer.

Recommended v1 resources

Use a small Python API and simple web interface on a **Cloud Run service**. Unlike a batch job, the service responds when a user asks a question. The recommended AI backend is **Gemini through Vertex AI**; choose its exact model and region during implementation. No AI resource has been provisioned yet.

Keep the assistant's hands small

- Allow approved, bounded read-only queries; do not let generated SQL write or delete data.
- Use the same release for knowledge and data. Include evidence with numeric answers.
- Test correct answers, empty results, conflicting sources and questions outside the dataset.
- Keep access controlled and put explicit limits on query size and model usage.

A good answer knows its limits
It can explain QC counts and table meanings. It cannot establish which disease a person has, infer unsupported cell labels, or make population claims from one donor.

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How we keep the factory reliable

Senior engineering is about proving that correct work stays correct when a run is repeated or interrupted.

Control	ELI5 explanation	What we do / will add
Checksum	A fingerprint for a file.	Reject a changed source or output instead of quietly accepting different bytes.
Release ID	A batch label for the recipe and ingredients.	Link source, policy, transformation code and runtime; keep different releases separate.
Safe replay	Press the button twice without making two copies.	Verify existing bytes/rows. GCS writes are create-only; BigQuery loading does not append duplicates.
Publish last	Put the 'ready' sign on only after inspection.	Upload and verify all release files before publishing the release manifest.
IAM identities	Different workers get different keys.	Separate workflow, QC, warehouse and assistant permissions; avoid committed credentials.
Audit + logs	A diary explaining who did what and when.	Keep structured platform logs plus durable run records. Add monitoring/alerts in operations work.
Recovery	Keep the evidence, repair, then retry.	Fail on corruption or conflicting data. Preserve suspect objects; do not bypass failed checks by changing a checksum.
Cost controls	Watch the meter and bound each task.	Limit workers and retries, cap BigQuery bytes per query, review storage versions, and verify budget alerts.

Current limits: QC uses 1 vCPU / 2 GiB with one retry; dbt uses two threads and a 100,000,000-byte billed limit per query. These are separate controls, not a monthly spending cap. Storage, query scans, execution time, logs, network reads and future model usage can all contribute to cost.

Known boundary: hard process termination can prevent the final application audit from being written. Check Cloud Run's platform execution status and logs as well. A workflow timeout does not automatically cancel an already-running job.

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What you will show a recruiter

A clear, reproducible story matters more than a long list of cloud products.

Demonstration	What it proves
1. Explain the diagram	You can connect a user's question to storage, transformations, orchestration and evidence.
2. Run one workflow	You can deploy and operate a pipeline using infrastructure as code and separate runtime identities.
3. Inspect the outputs	You preserve source identity, validate schema and counts, and reconcile warehouse results.
4. Run it again	You understand idempotency: unchanged release data and object versions, with a new audit for the new attempt.
5. Demonstrate a failure	You detect corrupt input or a failed test, stop downstream work, preserve evidence and recover safely.
6. Ask the assistant	You connect reliable data engineering to grounded AI without inventing biological facts.
7. Show tests + cost evidence	You make correctness and operating tradeoffs visible, rather than claiming unmeasured scale.

Final repository checklist

- A readable README with the problem, diagram, setup, example workflow and measured results.
- Versioned code/configuration, infrastructure, meaningful tests, dbt models and example OKF notes.
- CI checks on changes; approved deployment using short-lived cloud identity; no credentials in Git.
- A concise runbook, failure/recovery example, cost notes and a short recorded demonstration.

Verified evidence you already have
60 Python tests passed in the last full suite. The live BigQuery build passed **six models and 30 dbt tests**. Two direct Cloud Run executions and one workflow-triggered execution succeeded; verified replays preserved GCS object versions.

Step 7 should refresh this evidence after the remaining stages are implemented. Test counts are a snapshot, not a target to inflate. A claim belongs in the portfolio only when a reviewer can inspect how it was verified.

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A pocket glossary and reading list

You do not need to memorize the names. Remember the job each thing performs.

Term	Plain-language meaning
Ingestion / QC	Bring the data in / check it against explicit rules.
H5AD / Parquet	A scientific matrix-and-labels file / an efficient table file.
Sparse matrix	A grid stored efficiently because most entries are zero.
Schema / lineage	A table's structure / the trail showing where data came from.
dbt model / test	A saved SQL recipe / a rule checking that the results make sense.
Container / digest	A packaged program / the exact fingerprint of that package.
Job / service	A worker that finishes a task / an application that responds to requests.
Terraform state / IAM	The infrastructure inventory / rules for who may do what.
Idempotency / audit	Safe repetition / a record of an individual attempt.
OKF / grounding	Portable knowledge notes / basing an answer on specific evidence.
CI/CD / WIF	Automated checking and delivery / short-lived identity for external automation.

Authoritative reading

Public PBMC3k dataset and attribution

Scanpy PBMC3k loader

Cloud Storage

Cloud Run jobs

Workflows and Cloud Run jobs

BigQuery

dbt BigQuery configuration

Open Knowledge Format specification

Terraform GCS state backend

Workload Identity Federation

Evidence basis: repository source/configuration, docs/001 through docs/005, the BigQuery dbt run results, and the saved cloud/workflow verification reports. Snapshot date: 01 November 2026. Future services and implementation choices are labelled as planned or recommended; this guide does not provision them.